HTE staff checklist - Setup

Edition 2026-10-17 | Local OT-2 | 48 first-round + 23 second-round wells

48 pairs, then 22 pairs + L17 reference. Local OT-2; one 48 × 2 mL Para-Dox; **71 fresh reaction vials**. Individual screw caps by hand. No stir bars. Keep reaction and LC well maps distinct.

Before starting the clock

- Fit p300 single GEN2 left; p300 eight-channel GEN2 right. Reaction vials use single only (13 mm pitch). Check block/flat adapter/latch, offsets, cap clearance <=45 mm and tips >1 mm above bottoms. App-simulate released Python.
- Use released `dosing.csv`, `stock_preparation.csv`, `deck_loads.csv`, `lc_sequence.csv` and `round2_predose.csv`. Initial selection is prepared; software colleague handles local release/LC import. Stocks and initial scoring are outside the clock.
- Have 71 reaction vials/caps, 71 HPLC vials/caps plus calibration vials, two LC preparation plates, fresh robot racks and >=71 manual sample tips. Confirm full 1.7-min LC cycle and staged batch handoff.

Stocks - prepare before the run

Stock	Preparation / use
SM in DCM	**500 mM, 10 mL minimum**; 805.8 mg SM at 100% assay. Capped 20 mL glass bulk vial. Dose 50 µL, then evaporate completely.
Ligands / Pd	**40 mM in toluene**; individual masses/volumes in stock CSV, including second-round reserve and margin. Open/prepare ligands in glovebox. Keep bulk stocks capped; use ordinary plastic source racks.
Mesitylene	**100 mM in toluene, 5 mL**; 60.095 mg at 100% assay. Separate stock; add 25 µL AFTER DCM removal.
Quench	**9:1 ACN/water +0.15% v/v formic acid**, no IS. Prepare 100 mL; add 150 µL acid and make to volume with premixed solvent.
LC diluent	**1:1 water/ACN +0.1% v/v formic acid**, no IS. Prepare 125 mL; 125 µL acid, make to volume.

Correct masses for assay. Never load bulk preparation volumes into source tubes. Use **stage-specific deck loads**; refill for round two from reserves.

Slot	Contents
1 / 4	Ligand source racks; use exact CSV addresses, leave L07 absent.
2	A1 SM/DCM; A2 Pd; A3 toluene; A8 mesitylene. Quench A4/A6/A9; LC diluent A5/A7/A10/A11/A12.
3 / 10	Reaction block for pipetting / heater-shaker with flat adapter.
6	Standard 96-well LC preparation plate, seated throughout sampling.
5/8/9	Single tips / multi tips / multi tips. Slots 7 and 11 empty.

Calibration during first hold - 14 injections

For **each SM and product**: blank + **10, 50, 150, 350, 800 µM**, plus independent **666.7 µM** check. **Mesitylene fixed at 66.7 µM in every vial**, including blanks/checks. Use matched matrix and recipes in stock/calibration reference. Software colleague supplies calibrated method/channel and RT windows.

HTE staff checklist - Run

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Method / UV channel: _____ Operator / start: _____

Follow local App pauses. Keep capped vials capped until prompted to open. Record actual times and any deviations.

Step	Action
1. SM	Run released R1 dosing. Robot adds **50 µL 500 mM SM/DCM** to 48 vials. Move block off deck when prompted; evaporate DCM by HTEL's ventilated procedure. Cool/return to slot 3.
2. Assemble	Robot adds **25 µL mesitylene +37.5 µL ligand A +37.5 µL ligand B +75 µL Pd +75 µL toluene =250 µL**. Hand-tighten every cap; move/latch as prompted. Run **95 °C, 500 rpm, 4 h**.
3. During hold	Run calibration. Separately hand-dose **23 NEW labelled vials with 50 µL SM/DCM each**, using predose map in an ordinary off-deck rack. Evaporate completely, cap dry SM; add NO mesitylene/catalyst yet.
4. Cool	Move sealed block off heater as prompted. Cool before opening. Load released workup and matching sources.
5. Quench	Open caps for addition. Robot adds **500 µL quench per vial**. Recap ALL by hand. Move/latch and shake **1500 rpm, 2 min**, heater off. Return to slot 3.
6. Sample	At each batch pause, gently remix the indicated capped vials; settle bubbles, then uncap ONLY those targets. Robot prepares **980 µL LC diluent +20 µL extract**. Fresh sample tips; no bottom contact.
7. HPLC vials	At pause, recap sampled reaction vials. Keep preparation plate in slot 6. Fresh manual tip per sample: mix completed LC well **500 µL ×5**, avoiding air, then transfer **900 µL** to labelled HPLC vial. Screw cap; invert briefly. Start LC batch BEFORE resuming robot. Never take unfinished wells.
8. Decide	Complete all **48** injections. Software colleague joins exports, checks QC and generates **22 new pairs + L17 reference**. Archive first-round reaction vials. Fit 23 dry SM vials into SAME block; verify orientation.
9. R2	Refill from reserves using new deck loads. NO SM redosing. Add mesitylene/catalysts/toluene as selected; single reference uses **75 µL L17**, no ligand B. Hand-cap, run same 4 h, cool/work up/sample in batches of 8/8/7. LC preparation H3 is diluent only: DO NOT INJECT.

Stop clock after all **23** final injections and quantitative analysis. Archive CSVs, maps, QC, protocols and robot JSON logs.

Stop and correct

Visible layers, solids at aspiration, persistent bubbles, air aspiration or tip contact: pause/review. Repeated aspiration is not a solubility remedy. Reuse only approved same-stock/noncontact tips; samples always fresh. Uncertain transfer: fresh documented dilution, no guessed top-up.

Timing target: 717.7 /720 min, requiring 35 min LC/handling overlap. This is unmeasured with ~2 min margin; record real cycle/handling times. Details and local commands: reference handbook.